Region Proposal

Generating Candidate Object Locations

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1. What a Region Proposal Is

A region proposal is a candidate bounding box that **might** contain an object of interest, generated before full classification happens. Instead of exhaustively scanning every possible box location and scale in an image (computationally infeasible), a proposal step narrows millions of possibilities down to a few hundred or thousand promising candidates.

2. Why Proposals Exist

Classifying every possible sub-window of an image at every scale and aspect ratio is combinatorially expensive. Proposal methods act as a cheap first filter: high recall (don't miss real objects) matters more than precision at this stage, since a downstream classifier will refine and reject false positives.

3. Classical (Pre-Deep-Learning) Proposal Methods

- **Selective Search**: starts from an over-segmentation of the image (superpixels) and hierarchically merges regions based on color, texture, size, and shape similarity, producing boxes at many scales. Used by the original R-CNN.
- **Edge Boxes**: scores candidate boxes by how many contour edges are wholly enclosed inside them, on the intuition that objects tend to have well-defined boundaries.

Note: These methods are unsupervised, run on the CPU, and are not part of the neural network being trained - a key bottleneck they created.

4. The Region Proposal Network (RPN)

Faster R-CNN replaced external proposal algorithms with a small convolutional network - the RPN - that shares the backbone's feature map and is trained jointly with the detector.

- Slides a small network over the shared convolutional feature map.
- At each location, considers a set of **anchor boxes** - predefined boxes at multiple scales and aspect ratios centered at that point.
- For each anchor, predicts (a) an **objectness score** - is this likely foreground or background - and (b) **box regression offsets** to adjust the anchor into a tighter fit around an object.
- Applies Non-Max Suppression to the highest-scoring anchors to produce the final proposal set.

5. Anchor Boxes in Detail

Anchors are a fixed grid of reference boxes, typically combining several scales (e.g., 128, 256, 512 pixels) with several aspect ratios (e.g., 1:1, 1:2, 2:1). A feature map of size H x W with k anchors per location yields H*W*k candidate anchors, each independently scored.

6. Training the RPN

Each anchor is labeled positive or negative based on IoU overlap with ground-truth boxes:

- IoU > 0.7 with a ground-truth box (or the highest-IoU anchor for that box) - positive.
- IoU < 0.3 with all ground-truth boxes - negative.
- In between - ignored during training to avoid ambiguous supervision.

The loss combines classification (objectness) and regression (box coordinates) terms:

```
L = L_cls(objectness) + lambda * L_reg(box offsets) [only for positive anchors]
```

7. From Proposals to Final Detections

Proposals are not the final answer - they feed into a second stage (RoI Pooling/Align + a classifier and box regressor) that assigns a specific class label and further refines each box, discarding proposals that turn out to be background.

8. Proposal-Free Alternatives

One-stage detectors (YOLO, SSD, RetinaNet) and transformer-based detectors (DETR) skip an explicit proposal stage entirely, predicting boxes directly - trading the higher accuracy of region-based refinement for simpler pipelines and faster inference.

9. Key Takeaway

The shift from external, hand-crafted proposal algorithms (Selective Search) to a learned, end-to-end proposal network (RPN) was the key innovation that made Faster R-CNN both faster and more accurate than its predecessors, and it cemented 'proposal + refine' as a dominant paradigm in two-stage object detection.